EOS Composition Partials: Implementation Map v2

Current row-scoped EOS dxa, Skye speed path, Brunt, and implicit diffusion state

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This is the current implementation map for the development branch. It supersedes `notes/eos_composition_partials_implementat` as an architecture snapshot. The older file is still useful as the original plan.

Latest verification status: prior install/plotter validation exists from before the latest Skye Coulomb speed refactors. The latest edits in this map have had static checks only; MESA compile/model runs are user-owned.

Current Goal

The production goal is to make EOS composition partials available where the star solver actually consumes them, without replacing physics with limiters or finite-difference approximations.

The current production star-solver request is row-scoped:

$$\left\{\frac{\partial \ln P_{\rm gas}}{\partial X_j}, \frac{\partial \ln E}{\partial X_j}\right\}_{\rho,T},$$

with wider rows available for diagnostics and future consumers.

Primary Files

Area	Main files
EOS composition helper	`eos/private/eos_composition_partials.f90`
eosDT row/moment plumbing	`eos/private/eosdt_eval.f90`, `eos/public/eos_lib.f90`
Skye EOS partials	`eos/private/skye.f90`, `eos/private/skye_ideal.f90`, `eos/private/skye_coulomb.f90`, `eos/private/skye_composition_ad.f90`
Skye Coulomb leaves	`eos/private/skye_coulomb_liquid.f90`, `eos/private/skye_coulomb_solid.f90`
FreeEOS/OPAL table mapping	`eos/private/eosdt_eval.f90`
Timing diagnostics	`eos/private/eos_timing.f90`, `star/job/run_star_support.f90`
Star EOS calls	`star/private/eos_support.f90`, `star/private/micro.f90`
Implicit Brunt/Dmix	`star/private/implicit_brunt.f90`, `star/private/implicit_Dmix.f90`, `star/private/hydro_vars.f90`, `star/private/hydro_chem_eqns.f90`, `star/private/mix_info.f90`

Composition Gauge

MESA evolves constrained mass fractions:

$$\sum_i X_i = 1, \qquad \sum_i \delta X_i = 0.$$

Composition derivatives are therefore defined up to an additive constant:

$$\sum_i (Q_i + C)\delta X_i = \sum_i Q_i \delta X_i.$$

The star solver still applies its sink projection in `star/private/star_solver.f90`:

$$D_j^{\mathrm{sink}} Q = Q_j - Q_s.$$

The EOS side computes a consistent constrained isotope-column gauge from composition moments, then the star-side sink projection can be applied as usual.

Moment Derivatives

The shared helper provides the raw composition moment columns:

$$\bar{A} = \frac{\sum_i X_i}{\sum_i X_i/A_i}, \qquad \bar{Z} = \bar{A}\sum_i \frac{X_i}{A_i} Z_i,$$

$$Y_e = \sum_i X_i \frac{Z_i}{A_i}, \qquad m_c = \sum_i X_i \frac{W_i}{A_i}.$$

For a generic charge moment $M_g = \bar{A}\sum_i X_i g_i/A_i$,

$$\frac{\partial M_g}{\partial X_j} = \frac{\bar{A}}{A_j S_X}(g_j - M_g).$$

The active Skye number-fraction derivative is computed in the same helper. For active set A_{Skye},

$$y_i = \frac{X_i/A_i}{\sum_{a \in A_{\mathrm{Skye}}} X_a/A_a}.$$

The active-set derivative is used by both Skye ideal ions and Skye Coulomb composition partials so the two paths share the same constrained basis.

Row-Scoped EOS API

The internal EOS API has two modes:

Mode	Purpose	Rows
full dxa	diagnostics, plotter, finite-difference comparison	all `num_eos_basic_results`
row-scoped dxa	production star solve and Brunt faces	requested rows only

Production wrappers include:

Wrapper	Use
`get_eos_star_dxa`	cell-centered hydro rows
`get_eos_star_dxa_with_moments`	same, but reuses star-side moments

Wrapper	Use
`get_eos_brunt_dxa`	face/path Brunt rows
`get_eos_brunt_dxa_with_moments`	face/path Brunt rows with precomputed moments

The row-scoped path still writes into legacy `nv x species` buffers for stable indexing, but it avoids computing expensive component rows that were not requested. A future cleanup can carry smaller row-local buffers internally.

Legacy `eosDT_get` callers, including split burn and net one-zone burn, stay on `include_composition_partials=.false.` and do not enter the Skye/FreeEOS composition-partial path. They still need the two historical `d_dxa` rows, so the public wrappers keep a local full-row scratch buffer and copy those rows back without per-call heap allocation.

Skye Thermodynamic Packing

Skye remains a Helmholtz-free-energy path. For a composition derivative F_j, with T, ρ carried by `auto_diff_real_2var_order3`,

$$S_j = -\frac{\partial F_j}{\partial T}, \qquad P_{\mathrm{gas},j} = \rho^2 \frac{\partial F_j}{\partial \rho}, \qquad E_j = F_j + T S_j.$$

Then

$$\frac{\partial \ln E}{\partial X_j} = \frac{E_j}{E}, \qquad \frac{\partial \ln P_{\mathrm{gas}}}{\partial X_j} = \frac{P_{\mathrm{gas},j}}{P_{\mathrm{gas}}}.$$

`eos/private/skye_thermodynamics.f90:pack_composition_partials` computes only the requested rows. Derived rows such as `grad_ad`, `gamma1`, and heat capacity rows are available when requested but are not paid for in the normal hydro-row path.

Skye Ideal Ion Path

For

$$F_{\mathrm{ion}} = \frac{k_B T}{m_u \bar{A}} \Phi,$$

$$\Phi = \sum_i y_i \left[\ln\left(\frac{y_i n}{n_{Q,i}} \right) - 1 \right], \qquad n = \frac{\rho}{m_u \bar{A}},$$

the implemented derivative reuses the active number-fraction helper:

$$\Phi_j = \sum_i y_{i,j} \ln\left(\frac{y_i n}{n_{Q,i}} \right) - \frac{\bar{A}_j}{\bar{A}},$$

$$F_{\mathrm{ion},j} = \frac{k_B T}{m_u} \left(\frac{\Phi_j}{\bar{A}} - \frac{\Phi \bar{A}_j}{\bar{A}^2} \right).$$

The current speed refactor stores the repeated $\ln(y_i n / n_{Q,i})$ terms once per call and avoids rebuilding the full active-fraction Jacobian for every isotope column.

Skye Electron Path

Skye uses the HELM electron helper for the electron/positron free energy. With

$$D = Y_e \rho, \qquad F_e = Y_e f(T, D),$$

the composition derivative of each T, ρ coefficient is

$$\frac{\partial}{\partial X_j}\left[\frac{\partial^{a+b} F_e}{\partial T^a \partial \rho^b}\right] = Y_{e,j}\left[(b+1) Y_e^b f_{a,b} + \rho Y_e^{b+1} f_{a,b+1}\right].$$

The Skye wrapper maps this through $Y_{e,j}$ and fills electron/free-electron rows only when requested.

Skye Coulomb Path

Full Companion Path

The full companion path exists for general composition partials, phase rows, latent-heat rows, and phase-transition cells. It uses `skye_composition_ad_real`, which carries

$$(u, u_X),$$

where each entry is still a `auto_diff_real_2var_order3` in T, ρ. This lets the existing Coulomb formulas be copied closely while carrying a single composition perturbation.

It covers:

- liquid and solid OCP free energies;
- liquid and solid screening terms;
- liquid and solid mixing corrections;
- electron exchange-correlation;
- mixing entropy;
- Gamma-limit extrapolation;
- hard branch and soft phase transition.

Production Hydro-Row Basis

For ordinary hydro rows away from the phase transition, the current production path uses the basis

$$\{Y_e, \bar{A}, y_1, \dots, y_N\}.$$

Network isotope columns are assembled by

$$F_{C,j} = Y_{e,j} F_{C,Y_e} + \bar{A}_j F_{C,\bar{A}} + \sum_i y_{i,j} F_{C,y_i}.$$

This reduces the expensive Coulomb work from one companion call per network isotope to a small basis.

Off-Transition Hard-Branch Reuse

When the base Skye EOS returns exactly liquid or exactly solid and phase/latent composition rows are not requested, the hard-branch value is

$$F_C = F_{\text{branch}}(T, \rho, Y_e, \bar{A}, \{y_i\}).$$

In this branch:

$$x_{\text{nefer}} = N_A Y_e \rho,$$

so the Y_e basis derivative is exactly

$$F_{C,Y_e} = \frac{\partial F_C}{\partial \rho}\frac{\rho}{Y_e}.$$

The $\bar{A}$ basis derivative enters the Coulomb correction through the final unit conversion $kT/(\bar{A}m_u)$, so

$$F_{C,\bar{A}} = -\frac{F_C}{\bar{A}}.$$

These are algebraic reuses of the base Skye AD result; they are not physics approximations. The full companion path remains active for phase-transition cells and phase/latent rows.

Batched Active-Number-Fraction Path

The active-number-fraction basis uses `nonideal_corrections_dya`. For a single selected branch, each OCP leaf value

$$f_i(T,\rho)$$

is independent of the active number-fraction perturbation. Therefore the branch sum

$$F_{\mathrm{OCP}} = \sum_i y_i f_i$$

has

$$\frac{\partial F_{\mathrm{OCP}}}{\partial y_j} = f_j.$$

The current implementation caches the selected branch leaf values from the base Skye Coulomb evaluation and reuses them in the hard-branch batched `dYA` path. Near the phase transition it falls back to the full evaluation.

FreeEOS / OPAL / SCVH Table Components

FreeEOS and OPAL/SCVH naturally provide a compact table basis in X, Z. The current path keeps this compact pair through the component call and expands to isotope columns with the shared constrained helper.

This avoids doing the table work once per isotope. Timing from the 500-model benchmark showed FreeEOS dxa cost around 11.4 thread seconds, with table lookup around 2.6 thread seconds and `X,Z` expansion negligible.

eosDT Blends

The current production policy is fixed-weight composition rows through eosDT selector blends:

$$R_j = \alpha R_{1,j} + (1-\alpha)R_{2,j}.$$

The selector composition term

$$\alpha_j(R_1 - R_2)$$

is intentionally not used for production composition rows. This avoids adding composition derivatives of numerical EOS selector functions to the physical Jacobian. Some selector derivative plumbing still exists for diagnostics and future designs.

Long-term thermodynamic endpoint: if component EOSs can expose consistent potential-level information, the cleaner blend would blend a Helmholtz free energy first and derive P, E, S, χ, and composition rows from the single blended potential.

Brunt / Ledoux Path

The implicit Brunt path lives in `star/private/implicit_brunt.f90`.

With `implicit_diffusion_flag` and `use_Ledoux_criterion`, it computes the composition term from face EOS pressure-composition partials instead of from the old two-composition finite difference.

Schematically, the pressure-composition contribution is

$$\sum_i \left.\frac{\partial \ln P_{\rm gas}}{\partial X_i}\right|_{\rho,T} \Delta X_i.$$

The face EOS pressure-composition coefficient is treated as a current-iterate coefficient. Second derivatives of EOS composition partials are intentionally not part of this implementation slice.

When `use_Ledoux_criterion = .false.`, the implicit Brunt face EOS path is skipped and the composition term is kept zero. This is both physically consistent with the Ledoux-off choice and avoids paying face EOS dxa costs.

Implicit Diffusion Policy

The current split is

$$D_{\rm mix} = D_{\rm mix}^{\rm implicit} + D_{\rm mix}^{\rm explicit}.$$

In code, `D_mix(:)` is still the ordinary real MESA coefficient consumed by existing mixing and sigma routines. `Dmix_implicit(:)` carries the AD derivatives for the promoted component, and `Dmix_explicit(:)` is the real semi-implicit coefficient held fixed through Newton iterations.

Implicit component:

- MLT/TDC convective mixing in cells whose post-cleanup `mixing_type` from the last full `set_mixing_info` pass is convective;
- semiconvection in cells whose post-cleanup `mixing_type` from the last full `set_mixing_info` pass is semiconvective.

Explicit or semi-implicit component:

- thermohaline;
- overshoot;
- rotation;
- minimum mixing;
- `other_D_mix`;
- other post-processing or external hooks unless separately promoted.

The isotope residual uses the usual fixed-coefficient implicit diffusion form. During a full `set_mixing_info` pass, `set_Dmix_components(s,.true.)` refreshes both pieces directly: promoted MLT/TDC cells use `mlt_D_ad` in `Dmix_implicit`, and `Dmix_explicit` stores the non-implicit part of the current ordinary MESA coefficient. If the MESA total is smaller than the local MLT/TDC value, the promoted AD coefficient is scaled down so both stored components remain non-negative. The total coefficient is then rebuilt as `D_mix = Dmix_implicit%val + Dmix_explicit`, matching the full-pass MESA total.

During Newton iterations, `set_Dmix_components(s,.false.)` keeps `Dmix_explicit` from the last full `set_mixing_info` pass fixed and refreshes only `Dmix_implicit`. The promoted coefficient value and derivatives come from current `mlt_D_ad`, but the set of implicit zones follows the post-cleanup `mixing_type` from the last full mixing-info pass. This prevents the solver refresh from re-opening zones that full MESA cleanup converted to minimum, overshoot, rotation, or no mixing. The tradeoff is that a zone that becomes formally convective during a Newton iteration waits until the next full `set_mixing_info` pass before it can enter the implicit component. The solver refreshes scalar `sig(:)` values every implicit Newton iteration; nonlinear coefficient Jacobian terms are gated:

The sigma value always comes from total `D_mix`. The optional sigma derivatives are built from `Dmix_implicit`, so the Jacobian only differentiates the promoted implicit component.

Control	Default	Meaning
`implicit_diffusion_include_dsig_structure`	false	include structure derivatives of sigma
`implicit_diffusion_include_dsig_dxa`	false	include high-gain cross-species sigma derivatives

The `dxa` sigma block is only meaningful with Ledoux on, because its current source is the Ledoux composition contribution to Brunt.

Runtime Timing Snapshot

Recent user timing for a 500-model, no-implicit-diffusion benchmark with EOS composition partials enabled:

Quantity	Value
total wall time	30.715 s
EOS wall time	7.979 s
`thread_time_eos_dxa/threads`	7.721 s
`thread_eos_dxa_FreeEOS`	11.355 thread s
`thread_eos_dxa_Skye`	47.303 thread s
`thread_skye_dxa_ideal`	1.123 thread s
`thread_skye_dxa_coul`	16.700 thread s
`thread_skye_dxa_pack`	0.336 thread s

The main gain came from replacing redundant Skye Coulomb companion calls for `dYe` and `dabar` with algebraic reuse of the base AD result. A later speed refactor caches selected OCP leaf values for the hard-branch `dYA` path.

Verification Status

Completed earlier:

- install completed after the HELM `mu` derivative fix;
- EOS composition plotter produced finite-difference comparison outputs;
- static sign review of the implicit diffusion residual matched the existing `set_dxdt_mix` convention.

Still pending:

- compile/install after the latest Skye Coulomb speed refactors;
- focused finite-difference checks for the hard liquid, hard solid, extrapolated, and phase-transition Coulomb states;
- Brunt/Dmix runtime validation on the target stellar cases;
- HELM/FreeEOS derived-row audit;
- pure-PC internal composition derivative path, if it becomes a goal.

Open Follow-Ups

1. Keep the row-scoped API but eventually shrink internal `nv x species` scratch to row-sized blocks.
2. Add focused Skye Coulomb finite-difference tests for the new hard-branch reuse and OCP-cache path.
3. Decide whether selector composition derivatives should remain diagnostic only, or whether the project should move toward a potential-level blend.
4. Validate the implicit Brunt/Dmix path in the high-temperature burning cases that motivated the work.